

Jyothsna Karuparthi

karuparthi.jyothsna@gmail.com | [201-780-6856](tel:201-780-6856) | [linkedin.com/in/karuparthi-jyothsna/](https://www.linkedin.com/in/karuparthi-jyothsna/) | github.com/jyothsna-ssv | [Portfolio](#)

Software Development Engineer delivering impactful full-stack and backend solutions. Expertise: Java, Python, JavaScript, WebSockets, React, Node.js, and API design. Proven ability to scale systems, enhance data integrity, and fortify reliability through disciplined debugging and strong data modeling.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript, JavaScript

ML/AI: PyTorch, TensorFlow, LangChain, RAG, RLHF, Fine-tuning, NLP, MLflow.

Frontend: React, Streamlit, Vue.js, Next.js, Vercel

Backend : FastAPI, Flask, Django, SpringBoot, Socket.io, PostgreSQL, MySQL, Redis, MongoDB, REST, GraphQL

Cloud : Amazon Web Services (EC2, IAM,S3, Lambda, SQS, RDS, CloudWatch), Supabase(Auth, RLS, Edge Functions), CI/CD

EXPERIENCE

Software Engineer | MoneyGram | New Jersey, USA

Jul 2026 - Present

- Develop and maintain backend functionality for MoneyGram web portals, supporting payment acceptance and core transaction workflows.
- Diagnose and resolve portal issues, implementing backend fixes to keep existing features running reliably.
- Collaborate with engineering and client teams to clarify requirements, validate changes, and troubleshoot issues before they hit production.
- Support ongoing portal stability through backend updates and performance fixes as new requirements come in.

Software Engineer | Community Dreams Foundation | New Jersey, USA

Jul 2025 - Jul 2026

- Designed and launched a B2B commercial cleaning platform from concept to production, using React, TypeScript, Supabase, PostgreSQL, and Stripe across client, admin, and staff portals.
- Translated stakeholder requirements into product workflows and sprint priorities, coordinating development, business, QA, and security teams through the full product lifecycle.
- Engineered a hybrid pricing system, combining square footage tiers, frequency multipliers, industry adjustments, and admin overrides to automate quote generation, and built the backend on 15+ Supabase Edge Functions with RLS, JWT auth, and RBAC securing all three portals.
- Validated 18 end to end workflows and set up automated GitHub Actions deployment to DreamHost, supporting a production launch with zero critical incidents and 550+ visits from warm prospects.

Software Engineer Trainee | WebMD | Navi Mumbai, India

Apr 2023 - Jul 2023

- Completed WebMD's Helios full-stack engineering program, covering frontend development, REST services, Docker, Kubernetes, and CI/CD.
- Built an internal worklog calendar using Python, Vue.js, and PostgreSQL to centralize employee activity records and simplify daily logging.
- Added a manager-facing reporting interface for reviewing worklogs and tracking team progress across the organisation.
- Worked in an Agile team using Jira and Git, from task breakdown through code review and troubleshooting.

Software Engineer Intern | MinitzGo | Hyderabad, India

Aug 2022 - Mar 2023

- Built product discovery features for customers with React and TypeScript. Integrated paginated catalog APIs, added search filters, and enabled infinite scrolling so users could browse products without interruption.
- Set up real-time cart and inventory updates using WebSockets, so stock levels and basket changes appeared instantly without a page refresh.
- Improved authentication and checkout with form validation, route protection, and clearer API error handling for users.

PROJECTS

ArchAI : AWS Architecture Diagram Generator | React, TypeScript, Python, FastAPI, OpenAI API, Pydantic, Graphviz, Docker

- Built a full-stack app where a user describes their system in plain English (e.g. a web app with a load balancer, two EC2 instances, and an RDS database) and gets back a validated, downloadable AWS architecture diagram, no manual diagramming required.
- Designed a structured OpenAI pipeline using Pydantic schemas, service coverage checks, graph validation, and automated repair, so 100% of explicitly requested services make it into the diagram before rendering.
- Built a deterministic Graphviz renderer with an allowlisted service registry, label normalization, isolated execution, and subprocess timeouts, blocking arbitrary model-generated code from running.

ChopURL : Custom URL Shortener | Python, FastAPI, Redis, Vue 3, Supabase Auth, Docker, Vercel, Render

- Built and deployed a full-stack URL shortener with custom aliases, one-year TTLS, click analytics, and password reset, letting both anonymous and authenticated users create and manage shareable links.
- Designed Redis indexes, atomic alias reservations, click counters, pipelines, and sorted-set pagination to prevent collisions and lost increments; secured APIs with token validation, ownership checks, configurable rate limits, and per-user quotas.

CrowdShield : Real-Time Content Moderation System | Spring Boot, Redis, PostgreSQL, OpenAI, WebSockets

- Built an async queue and worker pipeline with Redis to decouple content moderation from user submissions, keeping classification reliable without blocking requests during traffic spikes.
- Integrated the OpenAI API with threshold-based scoring and retry logic to keep moderation outcomes consistent through transient failures.
- Designed PostgreSQL schemas for job states and audit logs, and built an admin dashboard with live WebSocket updates for reviewing flagged items and tuning moderation rules.

EDUCATION

New Jersey Institute of Technology

Master of Science (M.S.), Computer Science

New Jersey, USA

Jan 2024 - May 2025

Jawaharlal Nehru Technological University Hyderabad

Bachelor of Technology (B.Tech.), Computer Science

Hyderabad, India

June 2018 - Aug 2022

ACCOMPLISHMENTS

National Women Empowerment Award - AICTE LILAVATI

Apr 2021

Honored nationally for leading digital-literacy initiatives empowering 200+ women.

Academic Excellence & Leadership Distinction

Mar 2021

Honored for academic achievement and leadership in fostering innovation and student collaboration.